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ARTICLE

Thermodynamic analysis and optimization for steam methane reforming hydrogen production system using high temperature gas-cooled reactor pebble-bed module

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ABSTRACT

Thermodynamic analysis and optimization for the steam methane reforming (SMR) hydrogen production system using the high temperature gas-cooled reactor pebble-bed module power plant (HTR-PM) are investigated in this work. Based on the thermodynamic-equilibrium model, parameters of thermal efficiency (η), hydrogen production ($Y_{H2/methane}$), methane conversion rate (X_1) and carbon monoxide conversion rate (X_2) are calculated under the specified temperature (T), pressure (P) and water-to-carbon ratio (S). The influence of S, T, P on η, $Y_{H2/methane}$, X_1 and X_2 is then analyzed. It considers a wide range of operating conditions (T = 400–1200°C; P = 2–7 MPa and S = 2–10). The results show that the influence of on the system performance is significant. When T > 950°C, η and $Y_{H2/methane}$ increases slowly ($4 < S \leq 6$) or reduces ($S > 6$). For the operating conditions of HTR-PM (P = 7 MPa; S = 6 and T = 950°C), the maximum value of η is 63.44% and the maximum $Y_{H2/methane}$ is 3.3 mol. At last, system optimized parameters are illustrated.

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System thermal efficiency; Steam methane reforming; HTR-PM; Nuclear hydrogen production

1. Introduction

1.1 Nuclear hydrogen production

The main sources of hydrogen are from fossil fuels and water. Wind energy, solar energy, and nuclear energy are usually considered as the most promising candidates to provide heat for hydrogen production [1]. Since the nuclear energy does not generate any greenhouse gases and limited by geographical factors [2], over the last decades, various hydrogen production methods using nuclear energy as heat source have been studied and reported.

Among all the hydrogen production methods, thermochemical cycle (I-S cycle [3–7], Cu-Cl [8–13], Mg-Cl [14–17] and Co-Cl [18]), Hybrid Sulfur Cycle (HyS cycle) [19–22], high-temperature steam electrolysis (HTSE) [23–27] and steam methane reforming (SMR) [28–31] have attractive prospects. However, the thermochemical cycle, such as the HyS cycle and HTSE, is either new technology or not particularly efficient at hydrogen production [17,22,30,31]. In addition, SMR has been extensively applied in various industries and accumulated much practical engineering experience [32–35]. Approximately 96% of the hydrogen produced today are extracted from the reforming of fossil fuel feedstocks, with SMR accounting for 49% [36]. And some studies [37,38] reported that the cost of hydrogen production via SMR reduces 255 $/ton H_2 and decreases 8.8 kg CO_2 /GJ primary energy (natural gas fuel energy), when the heat is provided from the nuclear reactors instead of the combustion of the methane input. Thus, SMR using nuclear reactor as the heat source can produce hydrogen from the fully clean energy [39,40].

1.2 Thermodynamic analysis of hydrogen production via SMR

Many studies [41–45] about the thermodynamic analysis of SMR process under different operating conditions have been reported. Tabrizi et al. [46] performed a thermodynamic analysis of SMR by combining the method of Gibbs free energy minimization (GFEM) and statistical modeling under the conditions of the water-to-carbon ratio S = 0.5–3, pressure P = 0.1–5 MPa and reforming temperature T = 600–1200 K. Results showed that the temperature plays a crucial role in the parameters of methane conversion rate, selectivity of gas (H_2, CO) and H_2/CO ratio. Adiya et al. [47,48] studied four ways of SMR (conversional steam reforming (C-SR), sorption enhanced steam reforming (SE-SR), chemical looping steam reforming

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(CL-SR) and sorption enhanced chemical looping steam reforming (SE-CLSR) under the conditions of P = 1 bar, S = 3 and T = 600–750°C. Results showed that the optimal operation values of P and S for the four hydrogen production methods are atmosphere pressure and 3, respectively. The optimal temperature range for C-SR is 1000–1010 K. Wang et al. [49] investigated the steam reforming of methane (SRM) and dry methane reforming (DRM) by the combination of thermodynamics and regression analysis, and utilized the method of GFEM to clarify the effects of temperature, pressure, and feedstock ratio. Results showed that the increasing of temperature and feed composition are favorable for the reaction.

In addition, Pashchenko [50] performed a thermodynamic equilibrium analysis of SMR based on the method of GFEM in a wide range of operational conditions (T = 800–1400 K; S = 1–4; P = 10–20 bar). Results show that the increasing of water to carbon ratio and temperature have noteworthy influence on the methane conversion rate and hydrogen yield when T < 1200 K. Carapellucci and Giordano [51] developed a thermodynamic equilibrium model and investigated the effects of operating conditions on the performances of three reforming processes: steam methane reforming, dry methane reforming and autothermal methane reforming. It shows that the temperature corresponding to the maximum value of H_2 production reduces from 1100°C (S = 2, P = 3 MPa) to approximately 650°C (S = 5, P = 0.1 MPa) in the case of SMR. Meanwhile, the hydrogen production increases to 3.5 mol/mol CH_4.

Since the working pressure P is 7 MPa in the hydrogen production system with the High-Temperature Gas-Cooled Reactor Pebble-Bed Module power plant (HTR-PM), the range of the operating parameters (S < 5, P < 3 MPa) investigated in these studies are still not enough wide, which cannot be used to evaluate the performances of the SMR hydrogen production system using HTR-PM.

1.3 Nuclear hydrogen production using high temperature gas cooled reactor

Among the Generation IV nuclear reactors, the high-temperature gas cooled reactor (HTGR) including the HTR-PM reactor is considered as a favorable heat source for hydrogen production via SMR due to its high output temperature (750–950°C) [52–54]. Some studies [55–58] about the hydrogen production system of SMR using HTGR have been performed. The first integrated system of the nuclear hydrogen production via SMR was experimentally investigated in the 'EVA I and EVA II (Einzelrohr Versuchs Anlage)' project at the Nuclear Research Centre Julich [59–61]. The hydrogen production system of SMR connected to the HTTR (a HTGR with thermal output of 30 MW), was carried out to investigate the system performances of nuclear SMR at Japan Atomic Energy Research Institute (JAERI) [62–64]. Later, Yin et al. [65,66] investigated the thermodynamic analysis of the hydrogen production performance using the 10 MW High Temperature Gas-cooled Test Reactor (HTR-10). The simulation results show that the system performances (e.g. hydrogen production rate, thermal efficiency, etc.) are remarkably influenced by the inlet helium temperature. Chikazawa et al. [67] designed and evaluated a hydrogen production plant of steam methane reforming with Sodium Cooled Rapid Reactor (SCRP). And its economic potential is roughly performed. Wang et al. [39,68] investigated the thermodynamic performances of a coupling system that hydrogen production of steam-reforming for methane, ethanol or methanol using high-temperature helium for heating the feed gases. The results showed that the reforming temperature playing a vital role in the process. And thermal efficiency of the coupling system for methane, ethanol and methanol was 61%, 58.58% and 89.17%, respectively. Khorasanov et al. [69] investigated the feasibility of the nuclear hydrogen production system using BN-600 commercial fast reactor (sodium-cooled fast reactor) as heat source in Russia. And its prospects are impressive. Ghouse et al. [70] developed and validated a rigorous model for the integrated steam reforming and nuclear heat system based on first principles. Based on the Ghouse's work, Hoseinzade et al. [30,31] developed a dynamic and two-dimensional model to design a hydrogen production system via SMR using HTR-Module pebble bed reactor (60 MW) as heat source. In the past decades, some references [28,71] describing the integrated system of the SMR using nuclear reactor as the heat source have been reported. However, research with the HTR-PM (2x250 MWe) [29,72,73] reactor as the heat source to provide heat for SMR process has not been reported. Meanwhile, the operational parameters of the conversional SMR that have been reported are not suitable to evaluate the performances of the SMR hydrogen production system using HTR-PM. Therefore, there is a need to carry out the thermodynamic analysis and optimization for the SMR hydrogen production system using HTR-PM.

1.4 Scope of this paper

In this paper, Section 2 describes the mathematical model to analyze the thermodynamic behavior of the SMR hydrogen production system using HTR-PM. In Section 3, firstly, comprehensive analysis on the thermodynamic performances of SMR using HTR-PM are investigated. Then, theoretical models are developed to investigate the thermal efficiency (η), hydrogen production ($Y_{H2/methane}$), methane conversion rate

(X_1) and carbon monoxide conversion rate (X_2), etc. In addition, operation parameters for producing hydrogen using HTR-PM are obtained by optimization analysis. Section 4 comes to the meaningful conclusions.

2. Overview of system and model

2.1 SMR hydrogen production system using HTR-PM

SMR hydrogen production system using HTR-PM is developed to utilize the nuclear heat fully and achieve effective thermal efficiency (η) and hydrogen production ($Y_{H2/methane}$). The schematic flow diagram of the HTR-PM hydrogen production is described in Figure 1. Further, a mathematical model to analyze the thermodynamic behavior of this hydrogen production system is introduced.

Figure 1 shows the process flow chart of SMR hydrogen production system using HTR-PM, which can supply high-temperature heat of 2x250MWe with the temperature of T_0 to the intermediate heat exchanger (IHX) in the primary helium gas loop. Then IHX transfers the nuclear heat with T_1 to the secondary helium gas loop that is composed of a helium gas loop, a methane feed line and a steam feed line to heat the steam reformer (SR) and steam generator (ST), in which pressure is regarded as P_0 (7 MPa). High-temperature helium from IHX first flows to the reformer where the catalytic conversion of methane and water vapor generation occurs. During the heat transfer process, the temperature of the inlet helium drops from T_1 to T_2 (750–950°C) due to the loss of heat along the pipeline. ST is designed not only to fully utilize the residual heat of helium from the reforming reactor to supply steam for hydrogen production, but also to mitigate the thermal disturbance of the reactor and the fluctuation caused by changes of the secondary helium gas temperature and system pressure (7 MPa).

2.2 Reaction equilibrium model

There are two major chemical reactions considered in the reaction tube of the steam reformer: endothermic reaction of methane steam reforming, and exothermic reaction of water gas shift.

The steam reforming reaction:

$$CH_4 + H_2O = CO + 3H_2 \qquad \Delta H_1 = +205.8\text{KJ/mol} \tag{1}$$

The water gas shift reaction:

$$CO + H_2O = CO_2 + H_2 \qquad \Delta H_2 = -41.2\text{KJ/mol} \tag{2}$$

The overall chemical reaction is shown as below:

$$CH_4 + 2H_2O = CO_2 + 4H_2 \qquad \Delta H_3 = +164.6\text{KJ/mol} \tag{3}$$

The overall chemical reaction suggests that the steam methane reforming reaction is a process of endothermic and volume-increasing. Consequently, high-temperature and low-pressure are more beneficial to the process. The energy required to run the steam reforming process is provided by the HTR-PM that provides the outlet helium whose temperature is limited in a range of 750–950°C. Two thermodynamic models, the complete and equilibrium reaction model [65], are introduced in this work. However, since the complete reaction model particularly

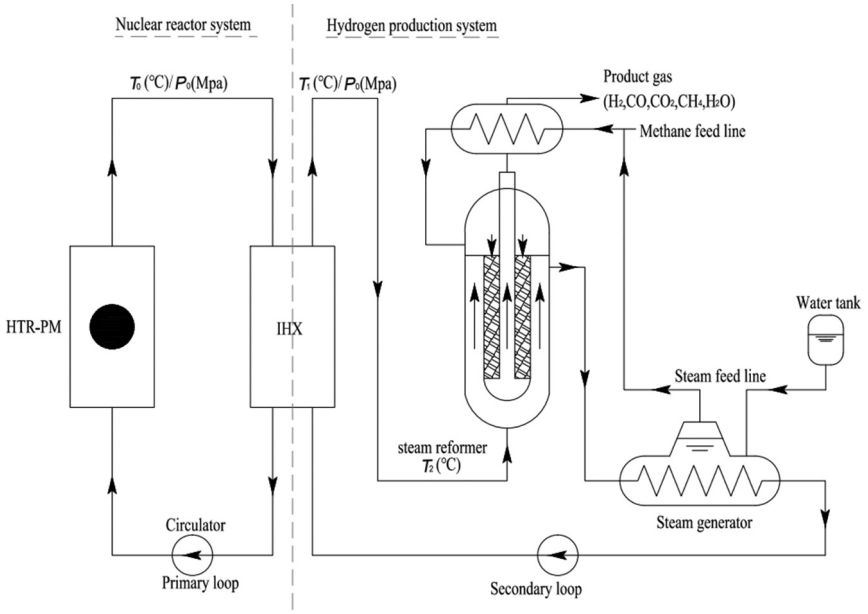


Figure 1. Schematic flow diagram of the HTR-PM hydrogen production.

considering the recovery of the latent heat is an ideal model, the equilibrium reaction model is used more frequently in the next sections. The development of equilibrium reaction model is shown as follow:

The flow diagram for substance flows in the SMR hydrogen production system using HTR-PM is shown in Figure 2. We assumed that the input feed is a mixture of methane and water. Steam reforming reaction and water gas shift reaction are only considered. The input heat is a sum of methane and Q_{in}, and Q_{in} is the heat of helium gas from the intermediate heat exchanger (IHX).

Methane conversion rate and carbon monoxide conversion rate [24] are determined as follow:

Methane conversion rate:

$$X_1 = \frac{P_{CO}^{outlet} + P_{CO_2}^{outlet}}{P_{CO}^{outlet} + P_{CO_2}^{outlet} + P_{CH_4}^{outlet}} \times 100\%, \quad (4)$$

Carbon monoxide conversion rate:

$$X_2 = \frac{P_{CO_2}^{outlet}}{P_{CO}^{outlet} + P_{CO_2}^{outlet}} \times 100\%, \quad (5)$$

where P_i^{out} is the partial pressure of species i at the out of steam reformer. Only reaction (1) and reaction (2) are taken into account as the model and the pressure difference is neglected. The equilibrium constants of both reaction (1) and reaction (2) are expressed as below:

$$K_{P1} = \frac{P_{CO}^{outlet} \cdot \left(P_{H_2}^{outlet}\right)^3}{P_{CH_4}^{outlet} \cdot P_{H_2O}^{outlet}} \times 100\%, \quad (6)$$

$$K_{P2} = \frac{P_{CO_2}^{outlet} \cdot P_{H_2}^{outlet}}{P_{CO}^{outlet} \cdot P_{H_2O}^{outlet}} \times 100\%, \quad (7)$$

where K_{P1}, K_{P2} are reaction equilibrium constants, which are listed in Table 1 [74].

The inlet gas composition of methane and water is defined as below:

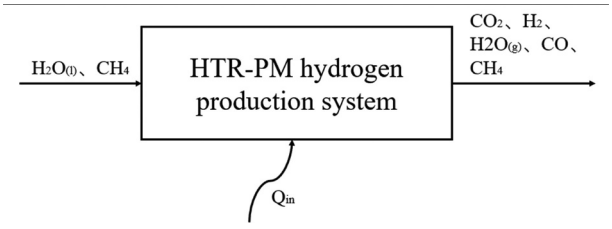


Figure 2. The flow diagram for substance flows in the HTR-PM hydrogen production system.

Table 1. Reaction equilibrium constants.

Reaction	Equilibrium Constants	Unit
Steam reforming	$1.198 \times 10^{11}\exp(-26830/T)$	$(MPa)^2$
Water gas shift	$1.767 \times 10^{-2}\exp(4400/T)$	$(MPa)^0$

$$S = \frac{F_{steam}^{inlet}}{F_{methane}^{inlet}}, \quad (8)$$

where F_{steam}^{inlet} and $F_{methane}^{inlet}$ are the inlet molar flow rate of water and methane, respectively.

Based on the thermodynamic-equilibrium theory and the reactions (1) – (2), and for the sake of simplicity, the initial methane gas is 1. The equilibrium gas composition can be obtained, which are shown in Table 2.

Then, based on the Table 2, the global reaction balance equation can be obtained:

$$\begin{aligned} &CH_4 + SH_2O_{(l)} = (3X_1 + X_1X_2)H_2 + (X_1 - X_1X_2) \\ &CO + X_1X_2CO_2 + (S - X_1 - X_1X_2)H2O(g) + \\ &(1 - X_1)CH_4 \end{aligned}, \quad (9)$$

According to the global reaction balance equation, the hydrogen production ($Y_{H_2/methane}$) is calculated by the following equation:

Hydrogen production:

$$Y_{H_2/methane} = (3X_1 + X_1X_2)F_{methane}^{inlet}, \quad (10)$$

where $F_{methane}^{inlet}$is the molar flow rate of methane.

According to the Dalton's law of partial pressure and substituting equilibrium constants (Table 1) into Eqs. (6) and (7), the functional relationship between S, T, P and X_1, X_2 can be deduced from the expressions of Eqs. (4) – (7), which are shown as follow:

$$\begin{aligned} &\frac{X_1^4(1 - X_2)(3 + X_2)^3}{(S - X_1 - X_1X_2)(1 - X_1)(2X_1 + S + 1)^2}P^2 \\ &- 1.198 \times 10^{23}\exp\left(-\frac{26830}{T}\right) = 0, \end{aligned} \quad (11)$$

$$\begin{aligned} &\frac{X_1X_2(3 + X_2)}{(S - X_1 - X_1X_2)(1 - X_1)} - 1.176 \times 10^{-2} \\ &\exp\left(-\frac{4400}{T}\right) = 0, \end{aligned} \quad (12)$$

Eqs. (11) and (12) are nonlinear equations. X_1 and X_2 are solved by MATLAB 2016b software under the specified S, T, P.

Hydrogen is the targeting product. Based on the Figure 2, the thermal efficiency is defined as follow [75]:

Table 2. Initial and equilibrium gas composition.

	Initial gas composition	generation	Consumption	equilibrium gas composition
CH_4	1	0	X_1	$1 - X_1$
H_2O	S	0	$X_1 + X_1X_2$	$S - X_1 - X_1X_2$
CO	0	X_1	X_1X_2	$X_1 - X_1X_2$
CO_2	0	X_1X_2	0	X_1X_2
H_2	0	$3X_1 + X_2$	0	$3X_1 + X_2$

$$\eta = \frac{n_{H2} HV_{H2}}{HV_{CH4} + Q_{in}}, \tag{13}$$

where HV_i is the lower heating value per mole of species i (i=CH_4, H_2). Q_{in} is the heat added to the system, which is provided by HTR-PM reactor. n_{H2} is the number of moles of hydrogen produced from one mole of methane. According to Eq. (9), n_{H2} is given as follow:

$$n_{H2} = 3X_1 + X_1X_2, \tag{14}$$

The combustion reaction of hydrogen is given as follow:

$$H_2 + \frac{1}{2}O_2 = H_2O_{(g)}, \tag{15}$$

And the lower heating value of hydrogen at standard state is defined as follow:

$$HV_{H2} = h^f_{H_2O(g)}, \tag{16}$$

where h^f_i is the formation enthalpy per mole of species i at the standard state. $h^f_{H_2}$ and $h^f_{O_2}$ are equal to zero at the standard state.

Similarly, a lower heating value of methane at standard state is defined as follow:

$$HV_{CH4} = h^f_{CO2} + 2h^f_{H_2O(g)} - h^f_{CH4}, \tag{17}$$

Q_{in} is the enthalpy change for the generalized reforming step, which is defined as follow:

$$Q_{in} = -\Delta H_R, \tag{18}$$

$$\Delta H_R = X_1X_2h^f_{CO2} + (X_1 - X_1X_2)h^f_{CO} + (S - X_1 - X_1X_2)h^f_{H_2O(g)} - Sh^f_{H_2O(l)} - X_1h^f_{CH4}, \tag{19}$$

Substituting Eq. (14) and Eqs. (16)-(19) into Eq. (13), and the thermal efficiency for equilibrium reaction model at standard state can be expressed as follow:

$$\eta = (X_1 + X_1X_2)h^f_{H2O(g)} / [(X_1X_2 - X_1)h^f_{CO(g)} + (1 - X_1X_2)h^f_{CO2(g)} + (2 + X_1 + X_1X_2 - S)h^f_{H2O(g)} - , (1 - X_1)h^f_{CH4(g)} + Sh^f_{H2O(l)}] \tag{20}$$

where h^f_i is the formation enthalpy of per mole of species i at the standard state. By substituting Eqs. (11) and (12) into the Eq. (20), the thermal efficiency is obtained.

3. Results and discussion

In this section, the influence of reforming temperature (T), water-to-carbon ratio (S) and pressure (P) on methane conversion rate (X_1) and carbon monoxide conversion rate (X_2) are investigated according to the two reactions of steam-reforming and water gas shift as given in Section 2. X_1 and X_2 also affect the thermal efficiency (η). Thus, the influence of X_1 and X_2 on η are also discussed. Further, the variations of thermal efficiency and hydrogen production ($Y_{H2/methane}$) with S, T and P are analyzed.

3.1 Analysis of methane and carbon monoxide rate conversions

Figure 3. shows the methane conversion rate (X_1) versus the reforming temperature (T) in the equilibrium reaction model (Eqs. (11)-(12)) for the pressure (P) of 7 MPa at different water-to-carbon ratios (S). X_1 increases rapidly with the increasing T since reaction (1) is a strong endothermic reaction. When $S > 2$, the maximum value of X_1 is 99.72% in a temperature range of 700–1000°C. In addition, it can be seen from Figure 3 that increasing S can promote X_1 when P and T are kept the same. This is because that increasing S leads reaction (1) forward to the positive direction. This relationship that X_1 increases with S at both the same pressure and temperature, which can be seen in Figure 4.

Figure 4 illustrates that methane conversion rate (X_1) versus steam-to-carbon ratio (S) for the temperature (T) of 850°C at different pressures (P) by the equilibrium reaction model. In general, X_1 gradually increases with S. When S = 10 and P = 7 MPa, X_1 reaches 92.1%. Besides, according to the simulation results from Figures 3 and 4, the increment of X_1 decreases as S increases. When $S < 6$, the increase of S by one can increase X_1 by 6%-12% for the pressure of 7 MPa as well as the temperature in a range of 700–950°C. When $S > 6$, increasing S by 1 only increases X_1 by 1%-3%. When $T > 950$°C and $S > 6$, there is little impact to increase S on X_1. Thus, increasing the vapor content of feed gases can appropriately improve X_1 when T and S are within

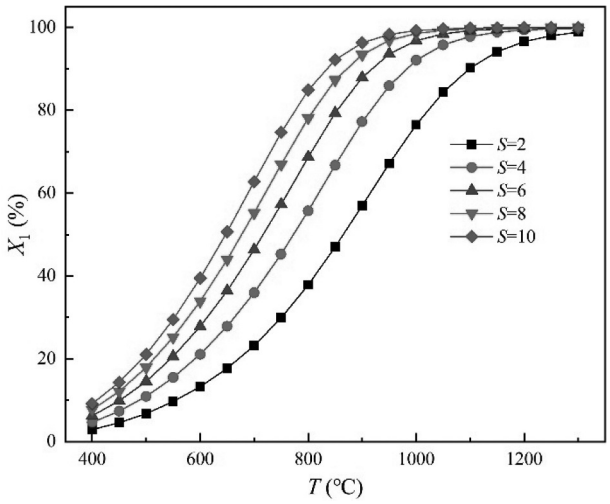


Figure 3. Methane conversion rate (X_1) versus reforming temperature (T) for the pressure (P) of 7 MPa at different water-to-carbon ratios (S) by the equilibrium reaction model.

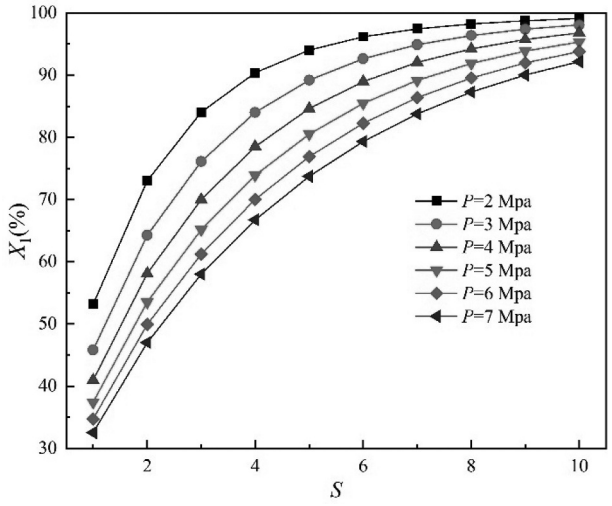


Figure 4. Methane conversion rate (X_1) versus water-to-carbon ratio (S) for the reforming temperature (T) of 850°C at different pressures (P) by the equilibrium reaction model.

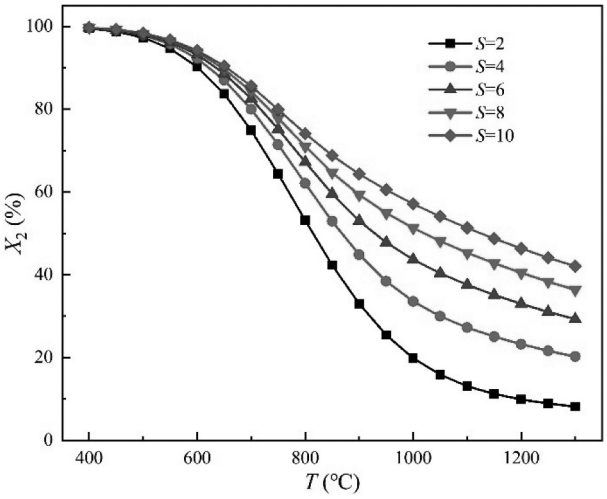


Figure 5. Carbon monoxide conversion rate (X_2) versus reforming temperature (T) for the pressure (P) of 7 MPa by the equilibrium reaction model.

a certain range for the SMR hydrogen production system using HTR-PM. Further, X_1 decreases with the increasing of P at the same S and T. This is because that the steam reforming reaction is a volume-increasing process as shown in reaction (1). However, P has little impact on X_1 according to Figure 4 and Figure 1s (supplementary materials). In addition, as shown in Figure 4 and Figure 1s, when $P > 3$ MPa, increasing P by 1 MPa decreases X_1 only by 1%-3% for $S = 6$ as well as in a temperature range of 700–950°C. Therefore, the influence of P on X_1 can be neglected for the SMR hydrogen production system using HTR-PM.

To verify the developed model, the results (solid line) are compared to the results (dash line) from Pashchenko's work [50]. All results are shown in Figures 2S and 3S, which indicate that the calculated results by our model are in excellent agreement with Pashchenko's results.

Figure 5 presents the carbon-monoxide conversion rate (X_2) versus reforming temperature (T) at different water-to-carbon ratios (S) for the pressure (P) of 7 MPa. As can be seen from this figure, X_2 is about 100% when $T < 500$°C. The variations of both T and S have no impact on X_2. When $T > 500$°C, X_2 decreases dramatically as P is 7 MPa. The relationship that the increase of T reduces X_2 is opposite to the relationship of X_1 and T (Figure 3). The first reason is that high temperature ($T > 500$°C) accelerates the reactive rate of reaction (1), and the second reason is that the water gas shift reaction (reaction (2)) is an exothermic process. The increase of T inhibits the conversion of CO into CO_2. However, when the temperature is more than 800°C, X_2 can be promoted by increasing S. Namely, the effect of S on X_2 becomes obvious when T increases at a certain pressure. Consequently, the effect of S should be considered for the SMR hydrogen production system using HTR-PM when $T > 700$°C.

Figure 6 illustrates that the carbon-monoxide conversion rate (X_2) gradually increases with the increase of water-to-carbon ratio (S) at different pressures (P) when the reforming temperature (T) is at 850°C. However, the process of water gas shift has an unfavorable tendency of X_2 as T increases. Hence, X_2 cannot be obtained as 100% in the current SMR hydrogen production system using HTR-PM. In addition, P does not affect X_2 because of the process of equal-volume, which can be seen from Eq. (2). Actually, X_2 increases slightly with the increase of P (Figure 6), which is also contrary to the results as observed in Figure 4. The reason is that the reactions (1)-(2) is a successive process and the slightly reduction of hydrogen production promotes X_2 as P increases. However, the increment of X_2 is so small that cannot compensate for the decrease of X_1 caused by the increasing of P. Therefore, SMR hydrogen production system using HTR-PM should be operated at high temperature and low pressure.

3.2 Thermal efficiency analysis of steam methane reforming

As can be seen from Eq. (11), the thermal efficiency (η) is related to the methane conversion rate (X_1) and the carbon-monoxide conversion rate (X_2). Figure 7 demonstrates the calculated results of η for the water-to-carbon ratio (S) of 4 and the pressure (P) of 7 MPa as well as the formation enthalpy at the standard state. Figure 7(a) shows that η is strongly dependent on X_1, and a linear increasing relationship is found between η and X_1. However, η is independent on X_2, as shown in Figure 7(b). Only when $X_1 > 0.4$, the influence of X_2 on η is slightly obvious as X_1 increases. Additionally,

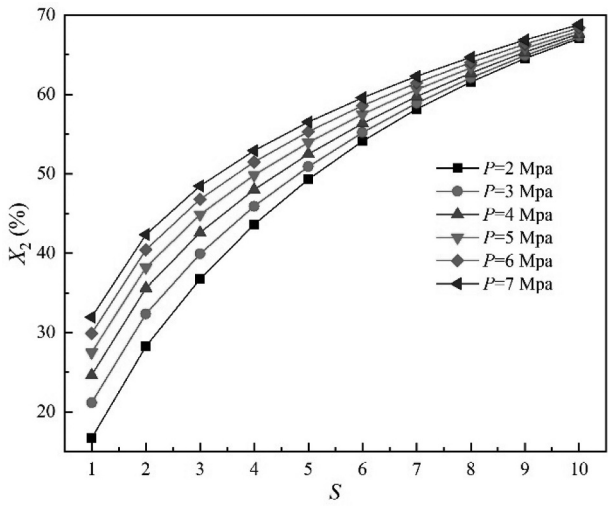


Figure 6. Carbon-monoxide conversion rate (X_2) obtained by the equilibrium reaction model versus water-to-carbon ratio (S) for the reforming temperature (T) of 850 °C at different pressures (P).

a three-dimensional graphic of η versus X_1 and X_2 is presented in Figure 7(c). As can be seen from this figure, the improvement of X_1 is one of the major ways to improve η for the SMR hydrogen production system using HTR-PM. In addition, it can be seen from Eqs. (11) and (12) that X_1 is also affected by S, T and P. Hence, the thermal efficiency and the hydrogen production ($Y_{H2/methane}$) are also investigated for different S, T, P in the following part.

Figure 8 presents the variation of thermal efficiency (η) with reforming temperature (T) for the pressure (P) of 7 MPa by the equilibrium reaction model. As can be seen from this figure, η significantly increases with T. However, η increases slowly ($2 < S \leq 6$) or reduces gradually ($S > 6$) when $T > 900$°C. The relationship between η and T is consistent to that between X_1 and T as shown in Figure 3. The reason is that high reforming temperature enhances X_1. And enhancing X_1 could further increase η (Figure 7(a)). It is noted that η begins to decrease slightly if T keeps increasing, which is due to the water gas shift reaction is an exothermic process and its reverse reaction rate increases as T grows up. The η reduces by 0.5%-1% as T increases every 100°C for the range that $S > 6$ and $T > 1000$°C. When $T = 1150$°C, $P = 7$ MPa and $S = 4$, the thermal efficiency can reach the maximum value of 66.3% theoretically. However, the reforming temperature (reaction (1)) is restricted in a range of 700–950°C as the limitation of the helium outlet temperature of HTR-PM reactor. When $T = 950$°C and $S = 6$, the maximum value of η for the SMR hydrogen production system using HTR-PM could only be 63.44% in reality. Figure 8 also shows the temperature that is corresponding to the maximum of η is reduced by adding water vapor into the feed gas. For example,

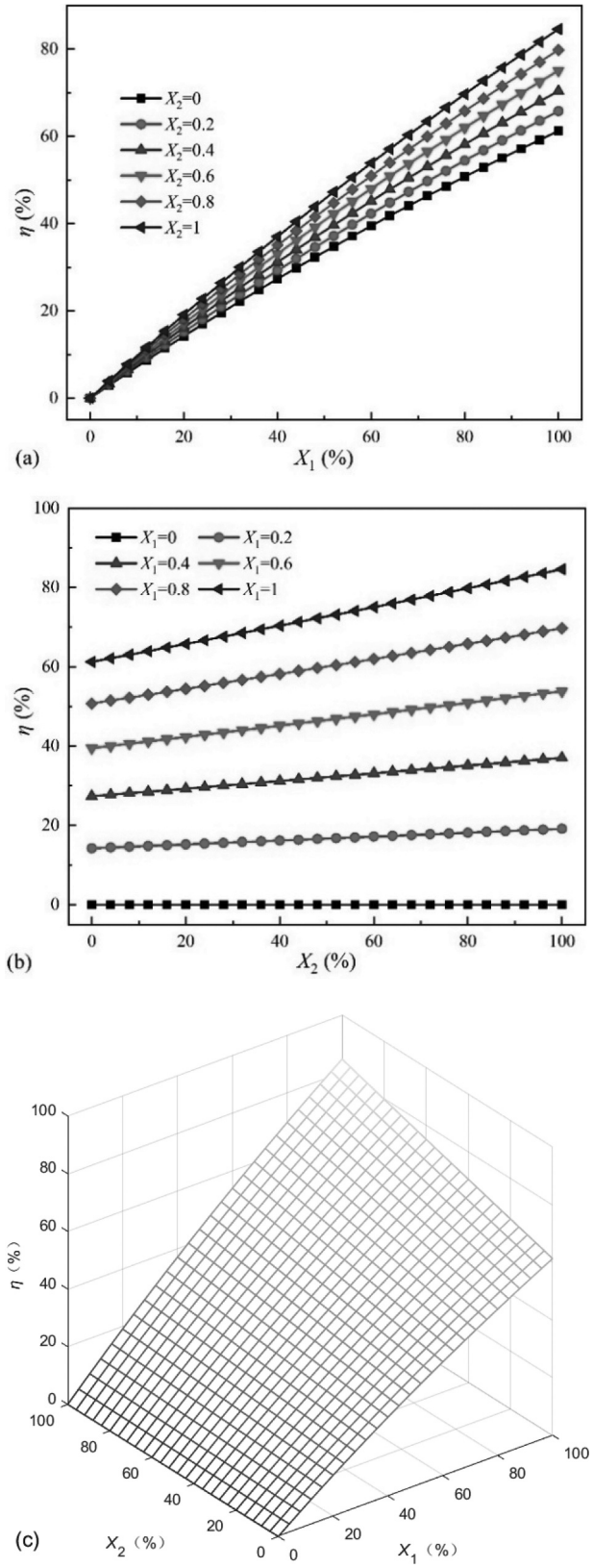


Figure 7. Calculated system thermal efficiency (η) of the hydrogen production system of HTR-PM when the water-to-carbon ratio (S) is 4 and the pressure (P) is 7 MPa. (a) system thermal efficiency (η) versus methane conversion rate (X_1) at different carbon-monoxide conversion rates (X_2); (b) system thermal efficiency (η) versus carbon-monoxide conversion rate (X_2) at different methane conversion rates (X_1); (c) system thermal efficiency (η) versus methane conversion rate (X_1) and carbon-monoxide conversion rate (X_2).

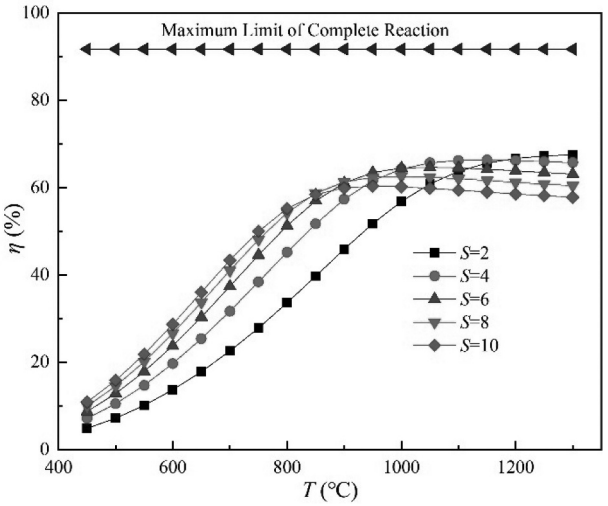


Figure 8. Variation of system thermal efficiency (η) with reforming temperature (t) at different water-to-carbon ratios (s) for the pressure (p) of 7MPa in equilibrium reaction model.

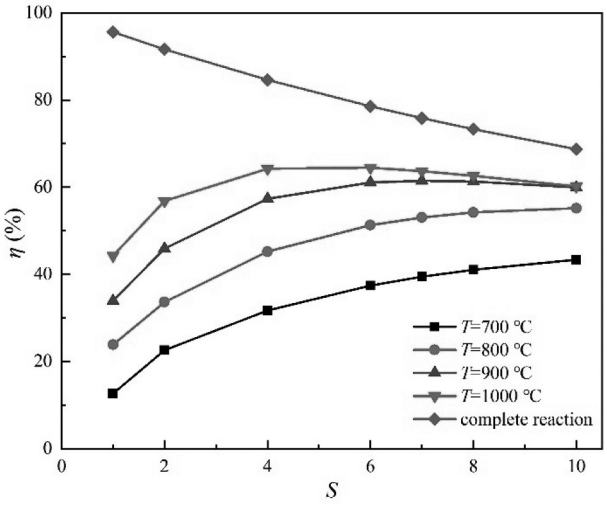


Figure 9. Variation of system thermal efficiency (η) versus water-to-carbon ratio (S) at different temperatures (T) for the pressure (P) of 7 MPa by the equilibrium reaction model.

when S = 4, the corresponding temperature of η is 1150°C. And the temperature changes to 1000°C when S increases to 6. Thus, T is the major factor to affect η. In a summary, when T < 950°C and $S \leq 6$, η can be enhanced by increasing T. And the temperature corresponding to the maximum value of η can be decreased by increasing S.

Figure 9 describes the variation of thermal efficiency (η) with water-to-carbon ratio (S) for the pressure (P) of 7 MPa at different reforming temperatures (T). In the complete reaction model, η monotonically decreases with S. However, in the equilibrium reaction model, a non-linear relationship exists between η and S. η increases gradually with the increase of S, then it begins to reduce slightly. The reason that η increases firstly with S is that the steam content in the feed gas cannot satisfy the requirements for steam methane reforming reaction when S is low (S < 8, T < 900°C), which results in a large amount of unreacted methane in the product gas. As shown in Figure 10, the molar fraction of methane at the reformer outlet decreases deeply with the increase of S. Then, the methane conversion rate (X_1) reaches to its maximum value as S increases. The molar fraction of methane decreases slowly. The molar volume fraction of methane is less than 0.61% when S > 10. This variation trend between the molar fraction of methane and S is consistent to that between X_1 and S (Figure 4). When η increases toward its maximum value, η exhibits a downward trend as S grows up. This is because that a significant amount of the unused steam takes the reaction heat out of the reformer and such heat cannot be recovered [65]. After η reaches its maximum value, η begins to decrease as T grows up. And the decreasing range of η becomes larger if T keeps increasing. As can be seen from this figure, the increase of S by one can reduce η by 0.5% after η reaches the maximum value of 63.44% for the temperature of 950°C. The increase of S by one

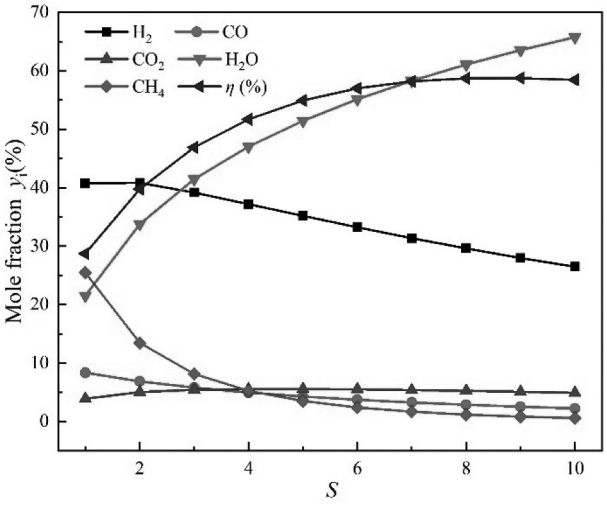


Figure 10. Mole fraction (y_i) of equilibrium composition with water-to-carbon ratio (<italic>S</italic>) for the reforming temperature (T) of 850 °C and pressure (P) of 7 MPa.

can reduce η by 1% after η increases to the maximum value of 64.78% for T = 1000°C. In addition, the relationship curves between η and S are plotted in Figure 4s in the supplementary material, which demonstrates the variation of η with S for T = 850°C as well as in different water-to-carbon ratios. The same conclusion also can be drawn from the Figure 4s. In a summary, S is another factor to affect η. However, its effect on η is less than T. Otherwise, the influence of S on η is complex. Only S within a certain temperature range can have a positive impact on η.

Figure 11 illustrates the relationship between the thermal efficiency (η) with pressure (P) for the reforming temperature (T) of 850°C at different water-to-carbon ratios (S) by the equilibrium reaction model. When S < 6, η decreases gradually with the increase of P. When $S \geq 6$, the increase of P by one reduces η by 1–

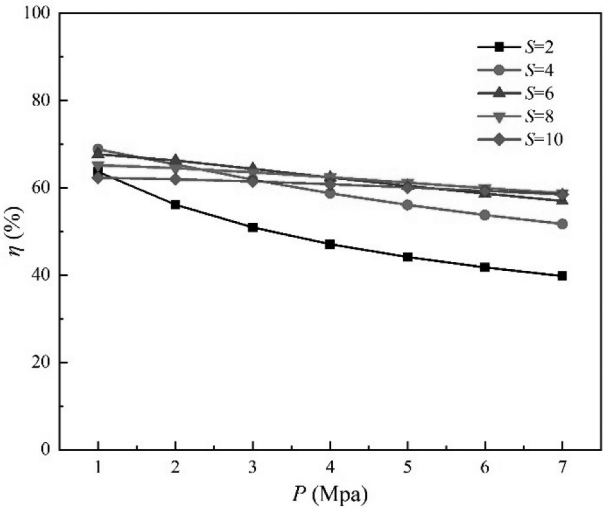


Figure 11. Variation of system thermal efficiency (η) with pressure (P) at different water-to-carbon ratios (S) for the temperature (T) of 850 °C in equilibrium reaction model.

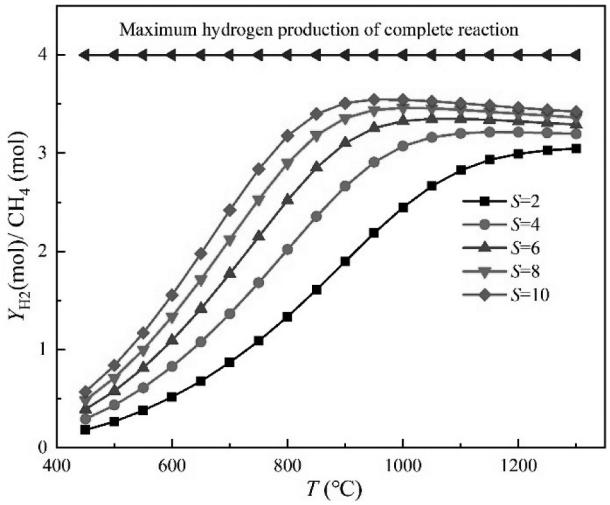


Figure 12. Hydrogen production ($Y_{H2/methane}$) versus the reforming temperature (T) at different water-to-carbon ratios (S) for the pressure (P) of 7 MPa by the equilibrium reaction model.

2%. And the reduction of η is not obvious as P increases, which can be attributed to the enhancement of both X_1 and X_2 which make up for the adverse effect of P on η. In a summary, when S is high ($S \geq 6$), P does not show obviously impact on η. Thus, in this situation, the effect of P on η can be neglected to analyze the performances of hydrogen production of HTR-PM.

3.3 Hydrogen production analysis

Hydrogen is the ultimate product gas needed to be collected after the reforming process. Therefore, hydrogen production ($Y_{H2/methane}$) is another critical parameter to evaluate the performances of the SMR hydrogen production system using HTR-PM. When X_1 and X_2 are equal to 1, the maximum of $Y_{H2/methane}$ is four times as much as the inlet mole flow rate of methane ($F^{inlet}_{methane}$) according to Eq. (10). Nevertheless, due to X_1 and X_2 show an opposite tendency as the temperature increases (Figures 3 and 5), X_1 and X_2 cannot simultaneously be equal to one. Figure 12 shows the $Y_{H2/methane}$ versus the reforming temperature (T) for the pressure (P) of 7 MPa at different water-to-carbon ratios (S) using the equilibrium reaction model. As can be seen from this figure, there is a nonlinear relationship between $Y_{H2/methane}$ and T. When T > 950°C, $Y_{H2/methane}$ increases slowly ($4 < S \leq 6$) or reduces ($S > 6$). When T = 1000°C and S = 6, X_1 is 96.83% (that is close to 100%) and $Y_{H2/methane}$ increases to the maximum value of 3.33 mol. The reason is that the positive reaction rate of reaction (1) is enhanced as temperature increases and a large amount of methane is converted into hydrogen. If T keeps increasing, $Y_{H2/methane}$ decreases slightly due to the reverse reaction rate of the water gas shift (reaction (2)) is strengthened. In addition, T corresponding to the maximum value of $Y_{H2/methane}$ is reduced by adding steam into the feed gas. For example, when S = 4, the temperature corresponding to the maximum value of $Y_{H2/methane}$ is 1150°C. And the temperature changes to 1050°C when S increases to 6. Thus, the decrease of T can appropriately improve the maximum value of $Y_{H2/methane}$ when S increases in the SMR hydrogen production system using HTR-PM.

Figure 13 presents the hydrogen production ($Y_{H2/methane}$) versus water-to-carbon ratio (S) for the temperature (T) of 850°C at different pressures by the equilibrium reaction model. $Y_{H2/methane}$ increases gradually with the increase of S, which is due to the forward reaction extent is enhanced as S increases. And methane could react fully with the water. In addition, the increase of S can always increase $Y_{H2/methane}$ rather than η. η increases initially with

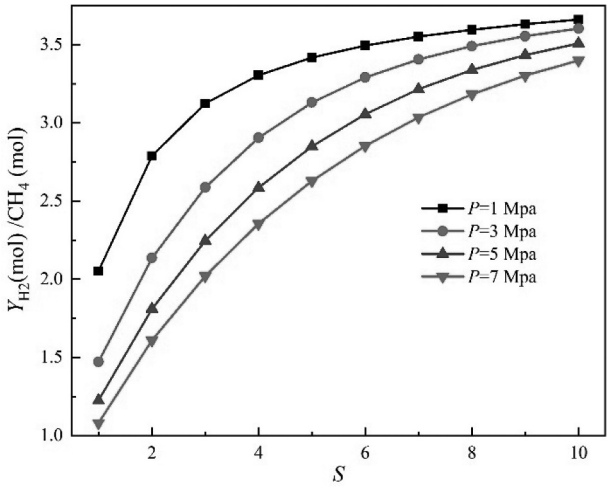


Figure 13. Hydrogen production ($Y_{H2/methane}$) versus water-to-carbon ratio (S) for the temperature (T) of 850 °C at different pressures (P) in equilibrium reaction model.

S and then reduces slightly. Thus, the synergistic influence of S on both $Y_{H2/methane}$ and η should be taken account into the design of the SMR hydrogen production system using HTR-PM. When T = 950°C and S = 6, $Y_{H2/methane}$ and η reach to 3.3 mol and 63.44%, respectively. It should be noted that an increase of P leads to a decrease of $Y_{H2/methane}$ when T and S are kept the same. However, the reduction of $Y_{H2/methane}$ caused by the increasing pressure is minor. When $S < 6$, $Y_{H2/methane}$ decreases by 0.1 mol as P increases every 1 MPa. When $S \geq 6$, $Y_{H2/methane}$ only decreases by 0.05 mol as P increases every 1 MPa. Thus, the effect of P on $Y_{H2/methane}$ can be neglected for the SMR hydrogen production system using HTR-PM.

3.4 The optimum parameters of HTR-PM hydrogen production

Finally, the performance parameters including the thermal efficiency (η), hydrogen production ($Y_{H2/methane}$), methane conversion rate (X_1) and carbon monoxide conversion rate (X_2) are calculated using the equilibrium reaction model for the SMR hydrogen production system using HTR-PM. Because the operating pressure (7 MPa) and the inlet helium temperature (950°C) are different from the data in reported references [39,60–66], the optimum parameters in the SMR hydrogen production system using HTR-PM are also different. Table 3 shows optimized parameters for η, $Y_{H2/methane}$, X_1, and X_2.

3.5 Comparison between HTR-PM and other reactors

Table 4 shows the performance parameters of other hydrogen production system in comparison to SMR combined with HTR-PM. The parameters include thermal efficiency (η), methane conversion rate (X_1) and carbon monoxide conversion rate (X_2). The first line refers to HTTR that is the mock-up test facility with a full-scale reformer tube designed in 1997 by JAERI. The design parameters of the reforming temperature (T), pressure (P), and the carbon-to-ratio (S) are 756°C, 4.2 MPa, 3.5 [64]. The second line refers to HTR-10 that had been analyzed in the work of [65], where T = 840°C, P = 3.2 MPa and S = 3–6. The last line shows the present study for HTR-PM, which sets the operation conditions as T = 850°C, P = 7 MPa and S = 6–8. This table provides a reference to design the hydrogen production system using different nuclear reactors.

Table 3. Optimized results of HTR-PM hydrogen production.

Parameters	Value
Range of HTR-PM operation parameters	
Reforming temperature (°C)	700–950
Reforming Pressure (MPa)	7
Water-carbon molar ratio	6–8
Calculation results	
X_1 (%)	46.35–96.86
X_2 (%)	47.77–82.50
η (%)	37.41–63.44
$Y_{H2/methane}$ (mol)	1.77–3.43

4. Conclusions

In this work, a thermodynamic analysis of methane steam reforming using HTR-PM as the heat source has been performed. The main conclusions are given as below:

Foremost, the temperature T is the most important factor that affects the hydrogen production performance. Namely, the thermal efficiency η, hydrogen production $Y_{H2/methane}$, methane conversion rate X_1 are highly related to T. Then, the influence of water-to-carbon ratio S on both η and $Y_{H2/methane}$ is small when the temperature is lower than 500°C. And the influence of S on both η and $Y_{H2/methane}$ become obvious if the temperature is larger than 500°C.

Meanwhile, η and $Y_{H2/methane}$ increases slowly ($4 < S \leq 6$) or decreases ($S > 6$) with $T > 950$°C. X_1 increases rapidly to its maximum value with the increase of T. It is close to 100% (96.83%) when T = 1000°C and S = 6. The effect of T on X_2 is also obvious, while an opposite trend is observed for carbon monoxide conversion rate (X_2). In addition, $Y_{H2/methane}$ increases as S increases. In terms of the operation parameters (P = 7 MPa, S = 6 and T = 950°C) for the SMR hydrogen production system using HTR-PM, the maximum value of $Y_{H2/methane}$ is 3.3 mol.

Furthermore, the calculated results show that η rapidly increases toward its maximum value with the increasing of S and then decreases. In this situation, η reaches its maximum value of 63.44% regarding to the maximum $Y_{H2/methane}$ of 3.3 mol. P does not have a significant impact on the system behavior (η, $Y_{H2/methane}$, X_1) when S is relatively high. Specifically,

Table 4. Theoretical or experimental data of different hydrogen production systems using nuclear energy.

	Heat source	X_1(%)	X_2(%)	η(%)
Experimental data	HTTR	54.8	62.5	30
Theoretical data	HTR-10	72.52–90.63	42.20–56.6	56.02–63.43
	HTR-PM	79.4–87.23	59.58–64.68	56.99–58.71

when $S \geq 6$, the effect of P on both η and $Y_{H2/methane}$ can be neglected for the SMR hydrogen production system using HTR-PM.

In general, thermal efficiency η, hydrogen production $Y_{H2/methane}$, methane conversion rate X_1 and carbon-monoxide conversion rate X_2 are firstly studied to analyze the system performances of hydrogen production using HTR-PM. A wide range of critical variables, such as water-to-carbon ratio (S=2–10), the reforming temperature ($T = 400$–1200°C) and pressure ($P = 2$–7 MPa), are discussed and analyzed by solving nonlinear equations in MATLAB 2016b software. Hydrogen production performances of different reactors are compared. Optimum parameters are illustrated. This work provides a realistic reference for the SMR hydrogen production system using HTR-PM.

Nomenclature

HTR-PM	The high temperature gas-cooled reactor pebble-bed module
JAERI	The Japan atomic energy research institute
HTTR	The high temperature engineering test reactor
HTGR	The high temperature gas cooled reactor
HTR-10	The high temperature gas-cooled test reactor of 10 MW
HTTR/SMR	The high temperature test reactor integrated with steam methane reforming
INET	The institute of nuclear and new energy technology
NFE	the project nuclear long-distance energy
SR	The steam reformer
ST	The steam generator
I-S cycle	The iodine-sulfur thermal chemical cycle
HyS cycle	The hybrid-sulfur thermal chemical cycle
EVAI	The einzelrohr versuchs anlage
HTSE	high temperature steam electrolysis
SMR	steam methane reforming
η	The thermal efficiency, %
X_1	The methane conversion rate, %
X_2	The carbon monoxide conversion rate, %
$Y_{H2/methane}$	The hydrogen production rate of one molar methane, mol
T	The reforming temperature, °C
P	The pressure, MPa
S	The water-to-carbon ratio
y_i	The mole fraction of i species
T_0	The outlet helium temperature of reactor, °C
T_1	The outlet helium temperature of intermediate heat exchanger, °C
T_2	The inlet process gas temperature at the catalyst bed, °C
P_0	The hydrogen production system pressure, MPa
P_i^{out}	The partial pressure of i species, MPa
K_{p1}, K_{p2}	The reaction equilibrium constants
$F_{methane}^{inlet}$	The methane molar flow rate of reformer inlet, mol
F_{steam}^{inlet}	The steam molar flow rate of reformer inlet, mol

Highlights

- The HTR-PM, a large commercial nuclear power plant demonstration reactor (2x250 MWe), is used to provide heat for hydrogen production of steam methane reforming. And its hydrogen production system performances are analyzed.
- A mathematical model to analyze the thermodynamic performance of the steam-reforming hydrogen production system using HTR-PM is proposed.
- System parameters related to hydrogen production efficiency are simulated by solving the thermodynamic equilibrium reaction model.
- The effects of the operation variables such as the reforming temperature, water-to-carbon ratio and the pressure on the system performance are analyzed.
- Operation parameters of the HTR-PM to achieve a high hydrogen production rate are optimized.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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